

APEX SOLUTIONS

Cyber Supply Chain Risk Management (C-SCRM) Framework

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Classification: Restricted - Operations & Logistics

1. PURPOSE & OBJECTIVE

The purpose of this procedure is to protect Apex Solutions' Repair-as-a-Service (RaaS) pipeline, wholesale operations, and B2B distribution contracts from hardware-based supply chain threats.

This framework establishes a strict verification protocol for all incoming bulk Apple parts, displays, top cases, batteries, logic boards, and chargers. Enforcing these controls mitigates the risk of introducing counterfeit goods, uncertified electronic waste, non-functional hardware, or malicious components into the enterprise Apple fleet lifecycle.

2. SCOPE & FRAMEWORK ALIGNMENT

This procedure applies to all incoming inventory lots received at the Apex Solutions processing headquarters in Rocklin, CA. It aligns explicitly with the following regulatory and industry standards:

- **NIST SP 800-161 Rev. 1:** Cybersecurity Supply Chain Risk Management Practices for Systems and Organizations.
- **R2v3 Standard (Responsible Recycling):** Focus on downstream tracking and component legitimacy.
- **Apple Certified/OEM Grading Metrics:** Ensuring A/B grade consistency for enterprise fleet deployment.

3. SUPPLIER RISK CLASSIFICATION MATRIX

To optimize the intake pipeline, Apex Solutions classifies component suppliers into three risk tiers. Technicians must check the master supplier registry before opening any incoming freight:

Supplier Tier	Description / Risk Profile	Required Inspection Level
Tier 1: Approved Premium	Apple Authorized, major carriers, or verified domestic corporate liquidators.	Standard Sampling: Check [REDACTED QUALITY CONTROL] of the lot.

Tier 2: Conditional	Secondary recommerce networks, brokers, or domestic wholesale liquidators.	Enhanced Sampling: Check [REDACTED QUALITY CONTROL] of the lot.
Tier 3: High Risk / Unverified	New international vendors, open-marketplace bulk lots (e.g., Alibaba, eBay lots).	100% Full Inspection: Every single part must be verified.

4. MULTI-PHASE PARTS INTAKE AND VERIFICATION PROCEDURE



(Diagram: Apex Solutions Multi-Phase Parts Intake and Verification Pipeline)

Phase 1: Material Intake Screening

1. Verify the delivery manifest against the physical package count.
2. Inspect the external shipping containers for tampering, water damage, or structural breaches.
3. Record the master Lot ID, Supplier Name, and date of arrival into the tracking system using [REDACTED DATALOGGER].

Phase 2: Physical & Branding Integrity Audit (Anti-Counterfeiting)

Technicians must carefully inspect individual parts for visual anomalies that indicate a component is a low-quality counterfeit or unauthorized clone:

- **Logo & Typography Check:** Inspect the component for authentic Apple markings, regulatory laser-etchings (CE, FCC), and crisp text. Blur-printed labels or mismatched fonts are instant indicators of counterfeit goods.
- **Structural Finish:** Check display assemblies and top cases for proper adhesive alignment, uniform glass sizing, and accurate flex cable placement.
- **Charger & Power Supply Specifics:** Inspect text on power adapters. Counterfeit Apple chargers often weigh significantly less than OEM adapters and exhibit uneven casing seams or spelling errors in the regulatory micro-print text.

Phase 3: Technical Functional & Diagnostic Testing

Before components are introduced to a client-owned device or an ITAD fleet refresh, they must be validated through electrical and digital diagnostics:

- **Batteries:** Plug incoming battery lots into automated diagnostic testing equipment: [REDACTED DIAGNOSTIC SOFTWARE]. Confirm that health percentages, design capacities, and charge cycles match the supplier's stated tier.
- **Displays & Logic Boards:** Mount the sample components onto dedicated testing benches. Verify correct color calibration, lack of dead pixels, and functional touch/True Tone controllers using [REDACTED FIRMWARE INTERFACE].

5. REJECTION, QUARANTINE, AND REMEDIATION PROTOCOLS

If a batch or an individual part fails any criteria within the verification checklist, the technician must execute the following containment isolation steps immediately:

1. **Immediate Isolation:** Stop processing the lot. Move all associated materials immediately to the Secured Quarantine Bin located away from active repair benches.
2. **Log the Infraction:** Complete an internal Supplier Non-Conformance Report detailing the exact failure vector (e.g., "Counterfeit labeling detected on Lot B-104").
3. **Chain of Custody Retention:** Keep the defective parts securely locked until the supplier issues a return authorization or Apex Solutions initiates an R2-compliant e-waste destruction routine to prevent the parts from re-entering the circular economy supply chain.

6. ROLES AND RESPONSIBILITIES

Apex Solutions enforces a strict separation of duties to ensure data protection, physical security, and regulatory compliance:

- **Compliance & Security Team:** Focuses entirely on policy creation, framework mapping, compliance tracking, and serving as the primary gatekeeper for formal R2v3, ISO, and enterprise client security audits.
- **Technical & Logistics Team:** [REDACTED PERSONNEL DETAILS]. Manages day-to-day hardware intake, executes technical repair routines, runs the data sanitization tools, and oversees the physical destruction of defective storage media.

- **Executive & Ownership Team:** [REDACTED PERSONNEL DETAILS]. Establishes long-term business goals, reviews supply chain risk metrics, manages client contracts, and authorizes budgets for security software and facility infrastructure.

APPENDIX A: APEX SOLUTIONS PARTS INTAKE VERIFICATION CHECKLIST

This operational artifact must be filled out by the processing technician for every incoming wholesale/parts batch.

- **Intake Date:** [REDACTED DATE]
- **Lot / Manifest ID:** [REDACTED COMPLIANCE RECORD]
- **Supplier Name:** [REDACTED SUPPLIER PROFILE]
- **Assigned Sourcing Risk Tier:** ☐ Tier 1 (Low) ☐ Tier 2 (Med) ☐ Tier 3 (High)
- **Component Type:** ☐ Displays ☐ Batteries ☐ Logic Boards ☐ Chargers ☐ Other

☐ PHASE 1: LOGISTICS CONTROL

- ☐ Box counts match the supplier invoice and delivery manifest.
- ☐ Shipping box seals are intact with no structural signs of external tampering.

☐ PHASE 2: VISUAL ANTI-COUNTERFEITING INSPECTION

- ☐ Hardware markings, serial formats, and Apple branding match OEM factory standards.
- ☐ Flex cables, solder points, and seams show zero signs of aftermarket modification or poor structural manufacturing.
- ☐ Font clarity and regulatory text markings are sharp, spelled correctly, and legally aligned with U.S. requirements.

☐ PHASE 3: TECHNICAL FUNCTIONAL TESTING

- ☐ Electrical testing via [REDACTED HARDWARE TOOL] confirms voltage stability and zero short circuits.
- ☐ Software/Diagnostic tools confirm correct internal microcode values and valid component metadata reporting.
- ☐ Battery testing indicates a health percentage above [REDACTED CRITERIA] and cycle counts under [REDACTED METRIC].

FINAL INTAKE DETERMINATION:

- ☐ **APPROVED:** Lot meets all criteria. Transferred to inventory storage code: [REDACTED SECURE STORAGE].
- ☐ **REJECTED:** Lot failed Phase [REDACTED STATUS]. Routed to locked Quarantine Bin code: [REDACTED LOCATION].